

Predicting Emergency Department Demand

NY facility-county-quarter panel • 2018–2024 • 57 facility counties • Total ED encounters



ROWS	COUNTIES	QUARTERS
1,536	57	28
facility county-quarter rows	facility counties	2018 Q1–2024 Q4

- Goal:** Quarterly ED-demand forecasting prototype.
- Target:** Total ED encounters aggregated by facility county and quarter.
- Decision:** Use previous-quarter persistence as the operational benchmark; keep XGBoost for diagnostics.



01 Data

Four sources were assembled into a county-quarter analysis panel for one-quarter-ahead forecasting.



PANEL	TARGET	GRAIN	PERIOD
1,536	ED	County-quarter	2018–2024
county-quarter rows	total encounters	facility county	28 quarters
DUPLICATES	TARGET MISS.	HOLDOUT	FEATURES
0	0	2024	20
county-quarter	total encounters	latest full year	selected predictors

Note:

Facility county is not always patient county of residence; this is a capacity/staffing prototype.

Data Inventory

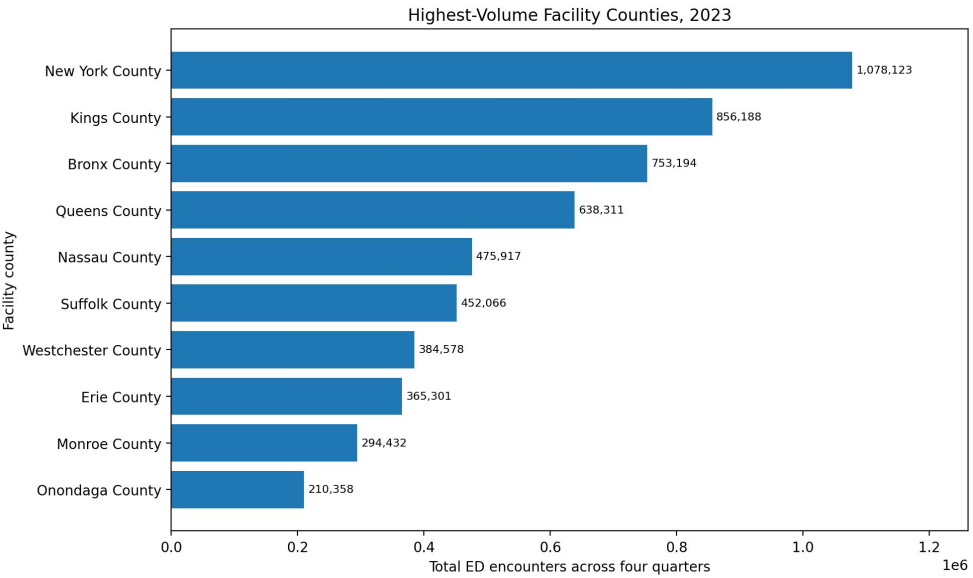
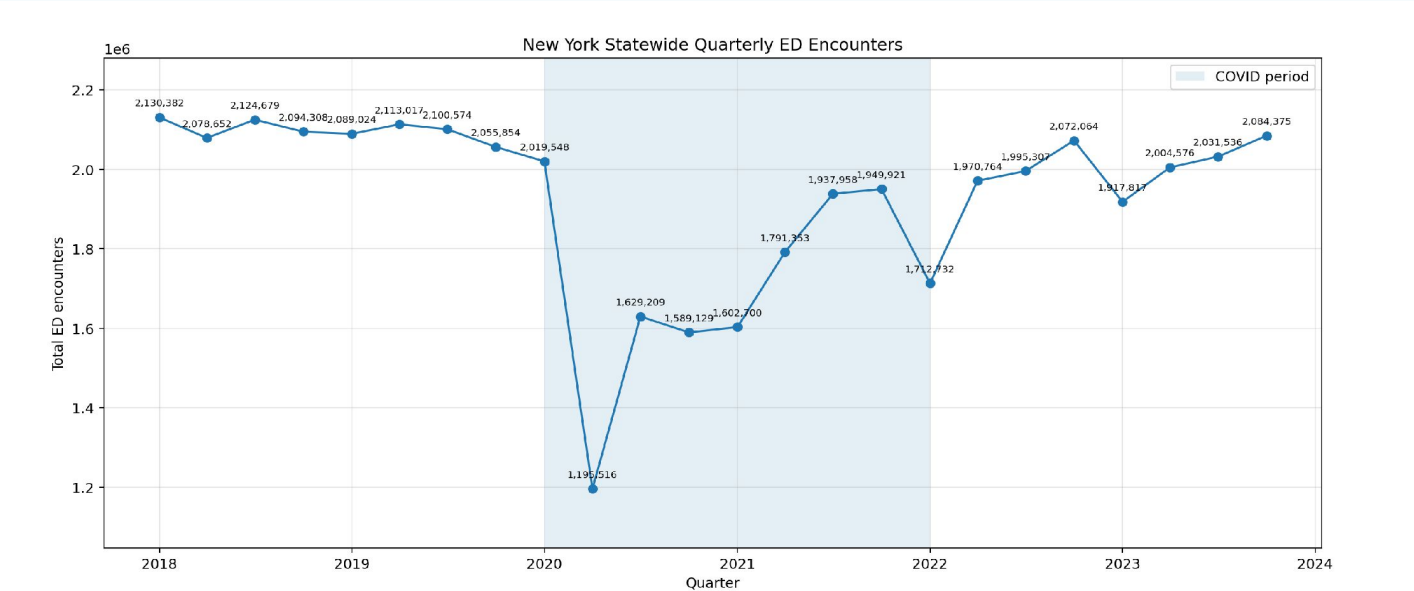
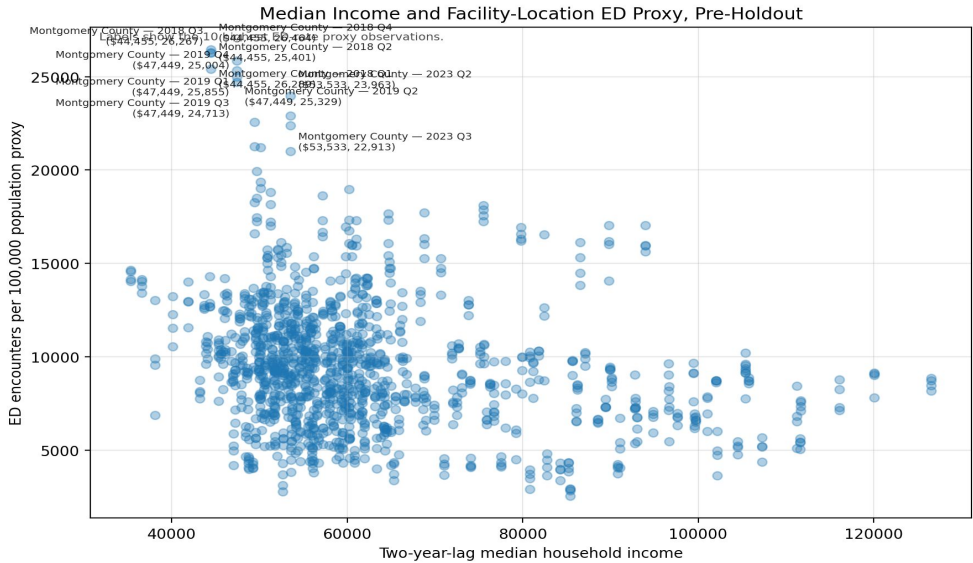
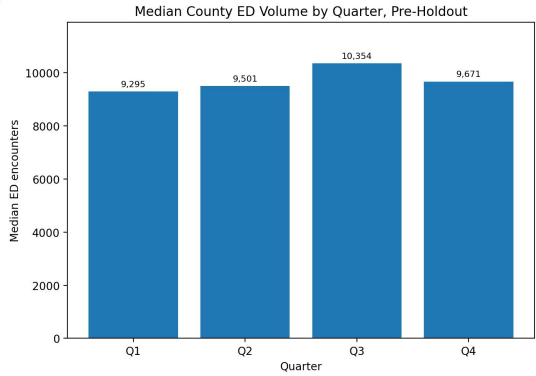
Source	Grain	Contribution	Status
SPARCS ED Encounters by Facility	facility-quarter	Forecast target: total ED encounters by facility county	Loaded + aggregated
U.S. Census ACS 5-Year	county-year	Lagged population, income, poverty, and age predictors	Loaded + lagged
Open-Meteo Historical Weather	county-day → county-quarter	Lagged weather totals, extremes, and seasonal conditions	Loaded + aggregated
Census Gazetteer	county	Latitude, longitude, and regional flags	Loaded + derived

02 EDA

COVID shock, recovery, volume concentration, seasonality.



HOLDOUT	2024 VOL.	HIGH QTR.	REGIONS
2024	7.91M	Q3	3
final year	ED encounters	median volume	NYC / downstate / upstate



EDA message	Operational implication
COVID caused a clear break; ED volume later recovered.	Use time-aware validation and a 2024 holdout, not random splitting.
Volume is concentrated in large facility counties.	Stakeholder review should focus on high-volume counties first.
Q3 has the highest median quarterly volume.	Seasonality matters, but lagged demand remains the strongest signal.

03 Features

Feature engineering links demand history, facility capacity, ACS context, lagged weather, and geography.

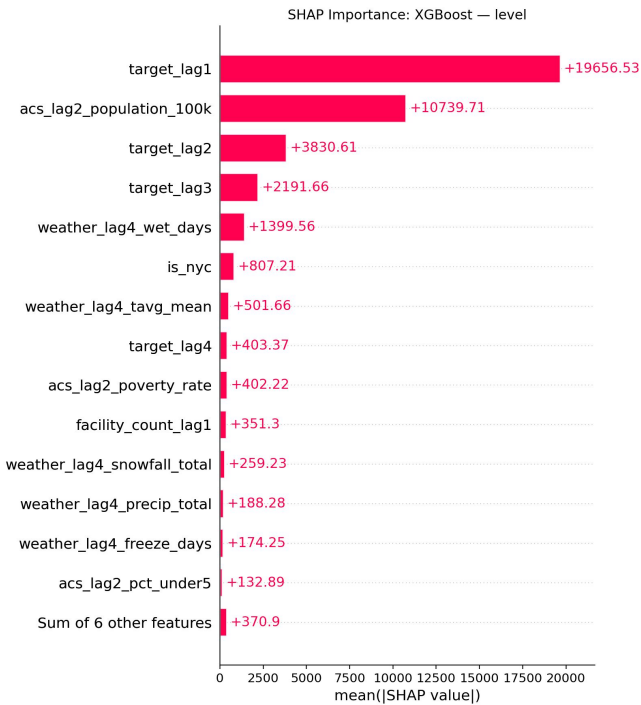
CANDIDATES	SELECTED	LAGS
25	20	1–4
screened features	final predictors	prior quarters

Feature Groups

Group	Updated feature role
Demand history	target_lag1, target_lag2, target_lag3, target_lag4
Facility capacity	facility_count_lag1
Socioeconomic	lagged population, income, poverty, age structure
Weather	lagged precipitation, snow, wet, hot, freeze days
Geography	lat, lon, NYC and downstate flags

Top selected features: target_lag1, target_lag2, target_lag3, target_lag4, acs_lag2_population_100k, facility_count_lag1

Leakage control: Expanding windows; ACS and weather predictors are lagged before forecasting.



04 Model

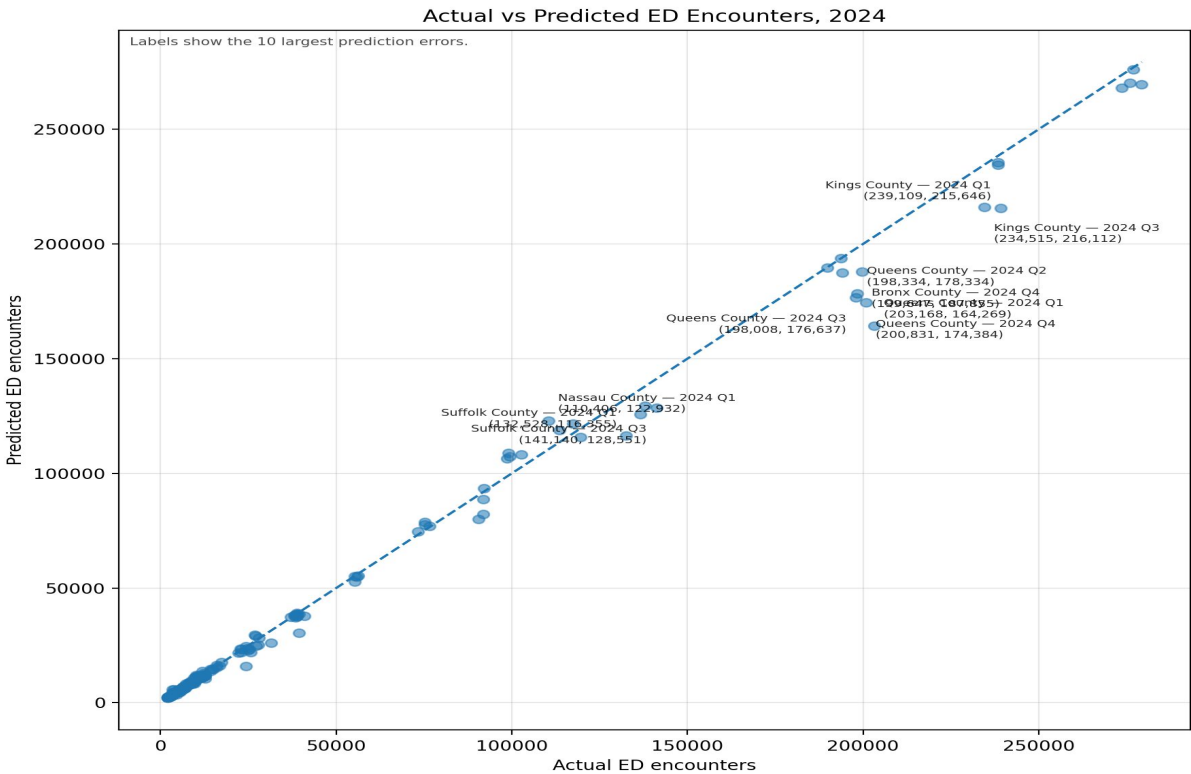
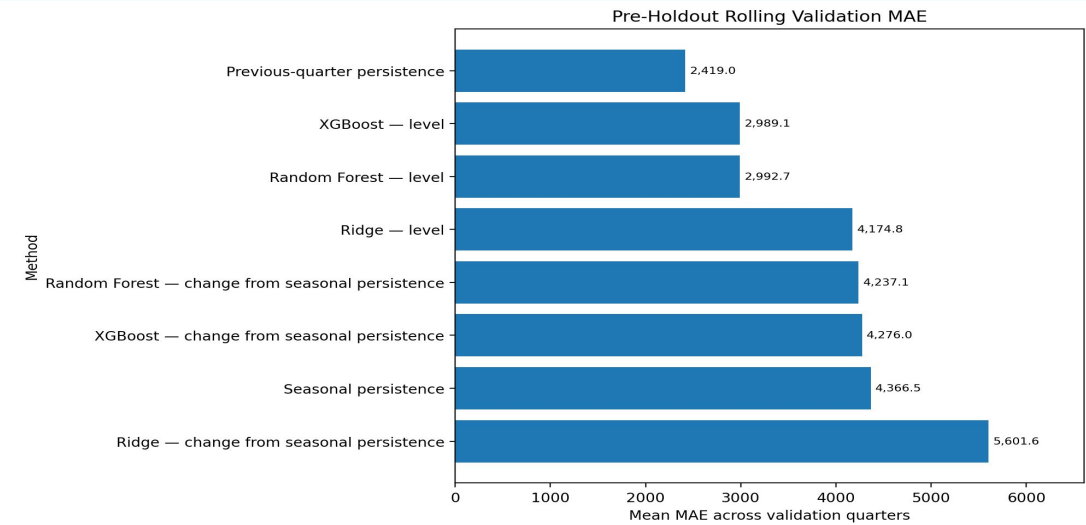
The tuned ML model is useful diagnostically, but previous-quarter persistence is the recommended prototype forecast.

Decision: Use persistence as the forecast shown to stakeholders; keep XGBoost to explain risk drivers and large errors.

BENCHMARK MAE	ML MAE	ML SKILL
1,253	2,158	-0.72
previous-quarter	XGBoost level	vs benchmark

Final 2024 Holdout Comparison

Method	MAE	WAPE	R²	Skill
Previous-quarter persistence	1,253	3.36%	0.9970	0.00
Tuned XGBoost — level	2,158	5.78%	0.9928	-0.72
Seasonal persistence	2,841	7.61%	0.9846	-1.27



Model message: A strong baseline is not a failure. It means quarterly ED volume is highly persistent, so the website should expose the prior-quarter benchmark clearly and use ML as a diagnostic layer.

05 Application Website

The deployed application helps hospital operations teams estimate next-quarter ED volume by facility county and review the result in context.



USER INPUTS	METHOD	OUTPUT
2	Persistence	Quarterly forecast
county + quarter	recommended method	planning context

Stakeholder Workflow

Step	What the website does	Stakeholder value
1. Select	Choose facility county and target period	Fast scenario lookup
2. Retrieve	Load recent demand and saved predictors automatically	No manual feature entry
3. Generate & Review	Generate the recommended forecast and show model evidence and recent context.	Support staffing and capacity discussion

Intended users

ED managers, hospital operations planners, staffing coordinators, and regional healthcare administrators.

Application purpose

Forecast next-quarter ED encounters by facility county to support staffing, shift coverage, bed, room, and capacity-planning discussions.

Deployment

Flask application hosted on Render, with code and model artifacts version-controlled on GitHub.

Live demo

Choose county → select target period → generate result
<https://ny-quarterly-ed-demand-app.onrender.com/>

Scan to open the app

